

MODULE-4

Construction and Demolition (C&D) Waste Management - Overview C&D Waste – Regulation, Beneficial Reuse of C&D Waste Materials.

Construction And Demolition Waste

- Construction and demolition waste means the waste comprising of building materials, debris and rubble resulting from construction, re-modelling, repair, and demolition of any civil structure.
- C&D waste includes bricks, tiles, stone, soil, rubble, plaster, drywall or gypsum board, wood, plumbing fixtures, non-hazardous insulating material, plastics, wallpaper, glass, metal (e.g., steel, aluminum), asphalt, etc., as shown in figure 1.
- However, C&D waste does not include any hazardous waste as defined under the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016.



Fig 1: Types of C& D Waste

Sources of waste generation in the construction industry:

C&D waste originates from diverse sources. Bulk generators include infrastructure projects (roads, bridges, flyovers) and real estate (housing, commercial buildings), producing large, uniform waste volumes. Small generators, like individual repairs or minor commercial works, contribute scattered, mixed debris.

The figure 2 shows the sources of C&D waste generated. Natural calamities—earthquakes, floods—generate sudden, massive waste, complicating management. The document notes demolition yields higher waste (300–500 kg/m³) than construction or repairs (40–60 kg/m³). This variability demands flexible strategies, from on-site reuse to centralized processing, tailored to source and scale.

Characteristics of the waste

C&D waste is an inert waste, over 90 per cent of which can be easily reused or recycled. A study by Technology Information, Forecasting and Assessment Council (TIFAC) in 2001 noted that C&D waste in India typically contains soil, sand and gravel (36 per cent), bricks and masonry (31 per cent), concretes (23 per cent), metal (5 per cent), wood (2 per cent), and others (3 per cent). But the composition of C&D waste varies from region to region, depending on prevalent construction style

and building material usage. There is no exhaustive study on regional variation in C&D waste composition. Few characteristics are mentioned below:

- **Composition:** C&D waste includes wood, concrete, steel, gypsum, masonry, plaster, metal, and asphalt.
- **Size and shape:** C&D waste components vary in size and shape.
- **Hazardous materials:** C&D waste can contain hazardous materials like asbestos and lead.
- **Heterogeneity:** C&D waste's diversity makes it difficult to segregate, recycle, and dispose of.

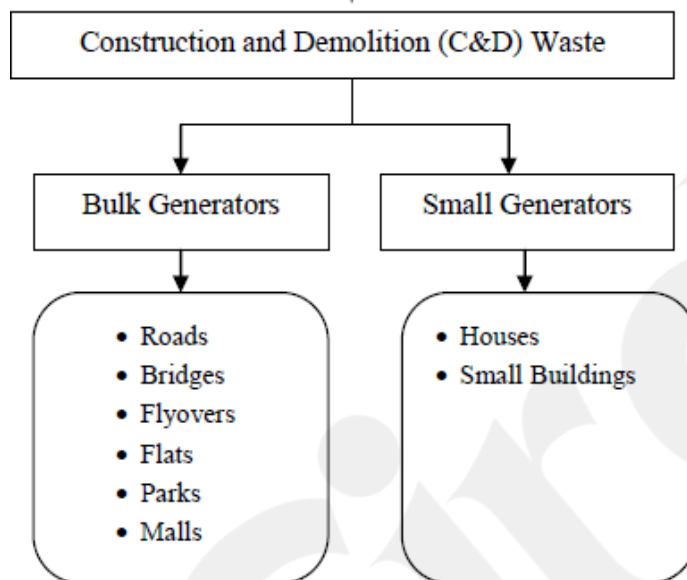


Fig-2: Sources of Generation of Construction and Demolition Waste

Quantum of waste generated: There is no scientific estimate available for the quantum of C&D waste generated in India. Over years different figures ranging from 10 million to 750 million tons per year have been given by various agencies (see *Table 1: Uncertainty in C&D waste estimates*). Official numbers are generally based on the assumption that about 25 per cent of municipal waste is C&D by nature, which in turn is based on typical characterization of municipal solid waste collected by cities. Given the reality that C&D waste collected by cities is accidental as it is not specifically collected and most of it is indiscriminately dumped in low lying land, the assumption of 25 per cent is a gross underestimation. In fact, most other nations report C&D waste as their biggest solid waste stream, many times their municipal waste generation.

Estimation of C&D waste generation

As part of its 2001 study, TIFAC developed the following rules of thumb to estimate C&D waste generation:

- Range 40–60 kg per sq. m of new construction,
- Range 40–50 kg per sq. m of building repair,
- Range 300–500 kg per sq. m for demolition of buildings.

Cities can use these rules of thumb in conjunction with their building/demolition permit data to arrive at a reasonable estimate of waste generated. But it must be kept in mind that a considerable part of this estimate, especially the demolition estimate, may be salvaged for reuse by the informal sector and may not require handling by the city.

Overview of Construction and Demolition Waste Management.

Construction and demolition (C&D) waste is complex due to the different types of building materials being used. In general, C&D waste may comprise the following materials:

- Major components** : Cement and concrete; bricks; cement plaster; steel (from reinforced concrete, door or window frames, roofing support, railings of staircase, etc.); rubble; stone (marble, granite, sand stone); timber or wood (especially demolition of old buildings)
- Minor components** : Conduits (iron, plastic); pipes (GI, iron, plastic); electrical fixtures (copper or aluminium wiring, wooden baton, Bakelite or plastic switches, wire insulation); panels (wooden, laminated); glazed tiles; glass panes; etc.

Asian countries

Asian institute of technology, Thailand had conducted a survey in various Asian countries and prepared a report regarding the construction and demolition waste management in May 2008. The study includes Asian countries like Bhutan, Japan, Hong-Kong SAR, China, Thailand and others including India. The following ie., chart shows the status of construction and demolition waste in Asian countries. Figure 3 shows the status of construction waste in Asian countries. and recycle (3R) practices in construction and demolition waste management in Asia', Thailand, May 2008. The Table 2 depicts Quantity of Waste Production by Developed Countries.

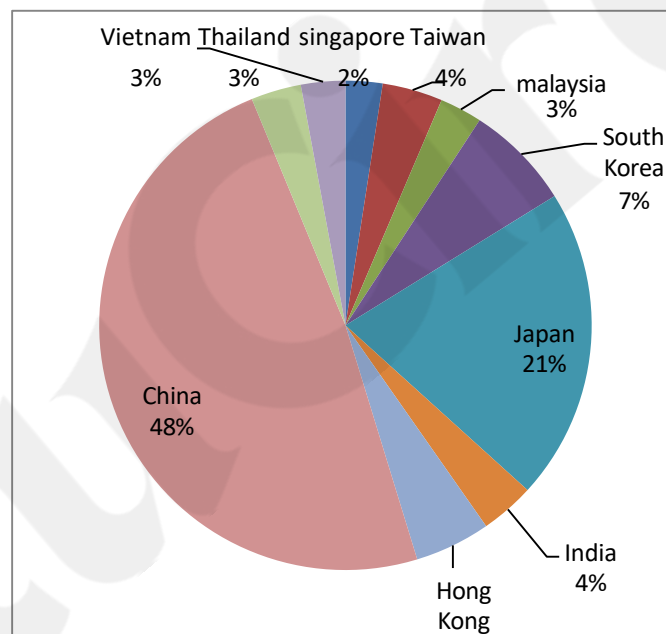


Fig- 3: Estimates of C&D Wastes in Some Asian countries

Table-2: Quantity of Waste Production by Developed Countries

Country	Amount Of Waste Per Year In MT	Year
Germany	223	2005
Australia	19	2008-09
China	200	2005
Japan	85	2000
	77	2012
South Korea	61.7	2013

INDIA:

There is no separate regulatory framework for handling the construction and demolition waste management in India, as it is considered in the municipal solid waste management. Due to which it is getting difficult to access the information or to handle the construction and demolition waste management. As report prepared by the MoEF (Ministry of Environment and Forest) in 2008 estimated that 0.53 million tons/day of waste is generated in the country. On that basis the 210 million tons of MSW is produced annually, table 1 shows the estimate prepared by central government of India. But as per the world bank report says Asian countries produces around about 1000kg per capita per year, it means the figure which stated by the MoEF is very less than the world bank report figure. This show in India is underestimating the construction and demolition waste handling. The figure 4 with graphical representation shows construction and demolition waste production per day in Indian cities.

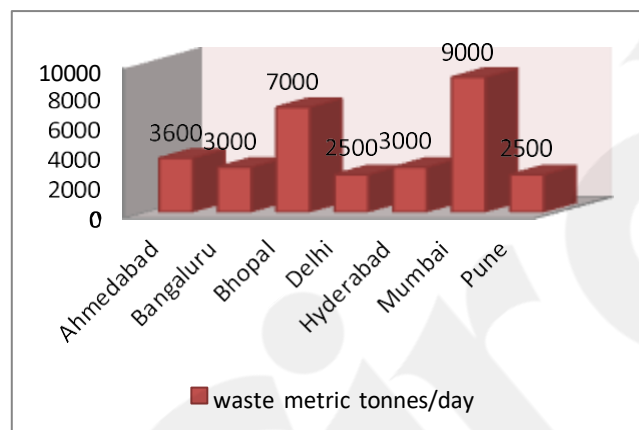


Fig -4: Waste Generated in Indian Cities Source

Composition of Construction Waste Generated in India

The composition construction waste depends on the type of structure as represented In table 3. For example, if the structure is flyover or bridge structures the composition will be usually concrete and steel. On the other hand, the if the residential structures are built or demolished the composition will be in verity, it consists of concrete, steel, wood, tiles, paints, plastics etc.

Table-3: Composition of Construction and Demolition Waste.

Components of C & D waste	FAC (%)	Survey (%)	Survey IL & FS (%)
Soil/sand, Gravel	36.0	43.0	31.5
Bitumen	2.0	-	-
Metals	5.0	-	0.4
Concrete	23.0	35.0	-
Wood	2.0	2.0	1.5
Others	1.0	1.0	7.6

Overview of C & D waste in Global and Indian context

- Globally, Construction and Demolition (C&D) waste is a major solid waste stream, accounting for 30-40% of total waste worldwide.
- In 2012, global C&D waste generation was approximately 1.3 billion tons, projected to rise to 2.2 billion tons by 2025 due to urbanization.

- The European Union produced over 820 million tons of C&D waste in 2014, with countries like Germany, France, and the UK as key contributors.
- In the United States, C&D waste reached 600 million tons in 2018, doubling municipal solid waste (MSW), with 75% recycled.
- The global push for a circular economy highlights C&D waste as a resource, promoting strategies like selective demolition and recycled material use.
- Challenges globally include inconsistent C&D waste definitions, limited reuse research beyond concrete, and environmental impacts from waste transport.
- In India, rapid urbanization drives a construction boom, increasing C&D waste, with the urban population expected to double to over 800 million by 2050.
- India's annual C&D waste estimates vary widely, from 12-15 million tons (official) to 150-530 million tons (research), reflecting data inconsistencies.
- India contributes 35-40% to global C&D waste, making it a significant producer due to infrastructure and housing demands (170 million houses by 2030).
- Over 90% of Indian C&D waste consists of bulky materials like soil, sand, gravel, bricks, masonry, and concrete, with the rest being wood, metal, and plastics.
- Despite high recyclability, India recycles only about 1% of its C&D waste, with most dumped in landfills, water bodies, or open spaces, causing pollution.
- The Construction and Demolition Waste Management Rules, 2016, mandate segregation, collection, and recycling in India, but implementation is slow.
- By 2020, only 13 of 53 targeted Indian cities had established C&D waste recycling facilities, falling short of expectations.
- India faces challenges like low builder awareness, weak enforcement, insufficient resources, and poor waste estimation and collection systems.
- Unlike advanced economies with strong recycling and policy frameworks, India struggles with systemic hurdles, contrasting with global circular economy trends.

Environmental impact due to C&D waste

Improper management of C&D waste impacts both the urban and natural environment in the following ways:

- C&D waste is routinely dumped in open drains and water channels, clogging them and leading to urban flooding during rains.
- Dumping of C&D waste in wetlands, water channels and riverbeds disrupts the hydrology and destroys the aquatic ecosystem.
- Leachate and fine chemical particles from C&D waste degrade the soil, leading to land and groundwater pollution. Especially hazardous components include paints, oil and asbestos sheets.
- C&D waste is filling up existing landfills and dump-yards, resulting in the need for more landfills or alternative dumpsites.
- C&D waste usually gets mixed up with other municipal solid waste during the process of transfer or at the collection site. Once mixed, it makes composting and/or recycling of MSW highly difficult.
- C&D waste includes hazardous substances such as sharps, broken glass, boulders, broken wooden logs, rusted metal, broken ceramics, etc., which create a hazardous environment when dumped on unfenced open places.
- C&D waste dumped on streets and footpaths blocks traffic and pedestrians, frequently contributing to traffic congestion and even accidents.
- C&D waste is one of the primary sources of fugitive dust pollution.

Not waste but a resource

India is undergoing a construction boom, which is leading to high rates of exploitation of primary

resources for construction materials. Sand, soil, stone, and limestone are critical resources used in the sector that are already facing supply disruptions and price spikes due to mining bans and restrictions. Therefore, use of secondary materials needs to be promoted to supplement the use of primary materials and recycled C&D waste is one of the best available options.

C&D Waste Management Rules, 2016

The Ministry of Environment, Forest and Climate Change (MoEF & CC) notified the Construction and Demolition Management Rules in 2016. This was the first ever of its kind and defines the responsibilities of each stakeholder.

Key highlights of the C&D Waste Management Rules, 2016:

- The Rules mandate use of recycled products in construction. Local bodies will have to utilize 10–20 per cent of material from C&D waste in municipal and government contracts for construction.
- The tipping fee for delivery of C&D waste to the recycling plant, and terms and conditions of civic body with the concessionaire may be designed to keep the price of C&D waste recycled products about 20 per cent lower than the corresponding conventional products.
- All large developers are accountable for collection and disposal of C&D waste.
- Local authorities to give appropriate incentives to waste generators for salvaging, processing, and recycling, preferably in situ.
- Recycling facilities will have to be created at a safe distance from habitation with adequate buffer zone.
- Local authorities to establish a database and update it once in a year.
- All renovation projects involving demolition, even in the private sector, may be mandated to use at least 20 per cent of recycled C&D waste products.

Under the C&D Waste Management Rules, 2016, Schedule I (14), the following are exempt from the norms of pollution from dust and noise:

- For construction work, where at least 80 per cent construction and demolition waste is recycled or
- Reused in-situ and sufficient buffer area is available to protect the surrounding habitation from any adverse impact.

Duties of waste generator

- Collection and storage of C&D waste generated within their premises.
- Ensuring that other solid waste is not mixed with C&D waste.
- Depositing C&D waste to collection centers or processing facilities as designated and authorized by local authority.
- Ensuring that there is no littering or deposition of C&D waste to prevent obstruction of traffic, public and drains.

Waste generators who generate more than 20 tons per day or 300 tons per project in a month shall:

- o Segregate waste into five streams: concrete, soil, metal, wood and plastic, bricks, and masonry.
- o Submit waste management plan and get approval from local authority before starting construction, demolition, or re-modelling work.
- o Pay relevant charges for collection transportation, processing and disposal as notified by local authority.

Duties of service providers and their contractors

- Prepare comprehensive C&D waste management plan for area within their jurisdiction.
- Clean C&D waste in the work area in a reasonable timeframe depending on the duration of work and quantity and type of waste generated. This should be done in consultation with local authority.
- Tie up with authorized agencies for cleaning of C&D waste if logistics support is not available.

Duties of local authority

- Issuing direction for management of C&D waste as per the Rules within their jurisdiction and seek detailed plan or undertaking as applicable from generators of C&D waste.
- Chalking out stages, methodology, equipment required, material involved in the activities required after Construction and Demolition.
- Safely disposing of C&D waste contaminated with hazardous, toxic or nuclear material after consultation with the concerned authority.
- Making arrangement for collection of C&D waste and ensuring that clean-up is done at regular intervals.
- Get the collected C&D waste transported to appropriate sites for disposal or processing.
- Giving incentives to generator for salvaging, processing and/or recycling C&D waste, preferably in situ.
- Examining and approve waste management plan of generators within one month or within date of submission and approval of building plan, whichever is earlier.
- Establishing C&D waste generation database and update once a year.
- Devising appropriate measures for management of C&D waste and use of recycled products in the best possible manner in consultation with expert institutions.
- Creating sustained system of IEC activities for C&D waste management through collaboration with expert institutes and civil society organizations and also disseminating through their own website.
- Providing incentive/s for use of products made with recycled C&D waste in construction activities.

Duties of State Pollution Control Boards/Committees

- Monitoring implementation of the Rules by local authorities.
- Giving authorization to C&D waste processing facility in Form III after examining the application in Form I.
- Preparing Annual Report with special emphasis on implementation status of compliance of the Rules and submit to CPCB before 31st July of each financial year.

Duties of state government or Union Territory administration

- Preparing policy document for management of C&D waste as per the Rules within one year of notification of the C&D Waste Rules. The responsibility for policy making lies with the Secretary in-charge of development in the state government or Union Territory.
- Providing land for storage, processing and recycling of C&D waste.
- Including site in approved land-use plan by Town and Country planning department.
- Making procurement of materials made with C&D waste up to a certain percentage mandatory in municipal and government contract, subject to strict quality control.

Duties of Central Pollution Control Board

- Preparing operational guidelines for management of C&D waste.

- Analyzing and collate the data received from SPCBs/SPCCs.
- Coordinating with SPCBs/SPCCs for any matter related to development of environmental standards.
- Submitting the Annual Compliance Report to the Central government before 30 August of each financial year based on reports given by SPCBs/SPCCs.

Duties of Bureau of Indian Standards and Indian Roads Congress

- Preparing code of practices and standards for use of recycled materials and products of C&D waste in construction and for roads.

Duties of Central government

- Facilitating local bodies in compliance of these Rules by MoUD, MoRD and MoPR.
- Reviewing implementation of the Rules when required by MoEFCC

Problems in recycling C & D Waste

- ☐ C&D recycling offers benefits, but challenges persist, as the U.S. recycles 455 millions of 600 million tons of C&D waste annually, leaving 145 million tons in landfills.
- ☐ Traditional demolition creates mixed rubble, making it hard to reclaim reusable materials, while even careful deconstruction requires early, organized separation to enable recycling.
- ☐ Contamination, such as wood or plastic mixing with concrete, reduces the quality of recyclable materials, complicating the process.
- ☐ The wide variety of C&D materials—concrete, wood, asphalt, metals, bricks, glass, plastics, and more—requires meticulous separation, increasing recycling challenges.
- ☐ Improper C&D recycling leads to financial costs by missing opportunities to reuse materials near construction sites, raising disposal and raw material extraction expenses.
- ☐ Environmentally, failing to recycle C&D waste causes soil degradation, deforestation, biodiversity loss, and community displacement from raw material extraction.
- ☐ Disposal of unrecycled C&D waste through landfills or incineration leaches toxic chemicals into water supplies, creating environmental hazards.
- ☐ Landfill use and improper disposal disproportionately harm vulnerable populations, contributing to environmental injustice.
- ☐ Challenges in C&D recycling, such as contamination and mixed rubble, stem from practical difficulties in demolition and deconstruction processes.
- ☐ Policy and operational improvements, like better on-site sorting and stricter regulations, could enhance C&D recycling rates and mitigate landfill reliance.
- ☐ Addressing these issues is crucial to reduce environmental impact, conserve resources, and promote a circular economy in construction.

Recycling of Construction and demolition (C&D) waste

Construction and demolition (C&D) waste is recycled using technologies such as sorting, crushing, and grinding. Other technologies include waste-to-energy conversion, robotics, and smart recycling bins.

The process of C&D recycling typically involves several steps:

1. **Collection and Transportation:** C&D waste is collected and transported to a recycling facility.
2. **Sorting and Separation:** Once the waste arrives at the recycling facility, it is sorted into different categories based.
3. **Crushing and Grinding:** Large pieces of concrete, masonry, and asphalt are often crushed and ground into smaller pieces for easier processing.
4. **Metal Separation:** Metals are separated from other materials using a magnet or eddy current separator.

5. **Wood Processing:** Wood waste is typically chipped or shredded into smaller pieces for use as fuel, mulch, or in the production of engineered wood products.
6. **Final Processing:** The separated materials are then processed further to produce new products or feedstock for the manufacturing of new products.
7. **Robotics:** Robots can disassemble electronic devices and sort component parts so that valuable materials can be extracted and repurposed.
8. **Waste-to-energy conversion:** Technologies such as incineration, gasification, and pyrolysis convert waste into energy.

C&D recycling systems generally consist of:

1. **Cranes for Pre-sorting:**
 - Cranes are used for initial pre-sorting at the recycling facility or demolition site, moving large volumes of C&D waste to separate oversized or hazardous materials before further processing.
 - Equipped with grapples or magnets, cranes lift and position waste, allowing operators to visually identify and remove items like large concrete blocks, steel beams, or debris that could damage downstream equipment or pose safety risks.
2. **Feeders:**
 - Feeders regulate the flow of C&D waste into the recycling system, ensuring a consistent and controlled supply to downstream equipment.
 - These can be vibrating feeders, belt feeders, or apron feeders, distributing waste evenly to prevent clogs or damage to shredders and screens. They handle bulk materials like rubble and debris, preparing them for size reduction or sorting.
3. **Size Reduction Equipment:** Reduces the size of C&D materials to make them manageable for sorting and processing into usable products, like aggregate or chips.
 - **Primary Shredders:** Large, heavy-duty machines (e.g., jaw crushers or impactors) that break down bulky C&D waste, such as concrete, asphalt, or wood, into smaller pieces (typically 6-12 inches). They handle initial crushing of mixed rubble, preparing it for further processing.
 - **Secondary Shredders:** Finer shredders (e.g., hammer mills or granulators) process pre-shredded material into smaller, uniform sizes (e.g., 1-3 inches) for specific recycling applications, like producing fine aggregate or wood chips. They refine material after primary shredding to meet quality standards.
4. **Screening Systems:** Separate materials by size to facilitate further sorting and processing, ensuring uniformity for recycling.
 - **Trommel Screens:** Rotating cylindrical screens with perforations that sift materials by size, allowing smaller particles (e.g., sand, gravel) to pass through while larger items (e.g., concrete chunks) move to the next stage. They're effective for wet or sticky C&D waste but can clog with fines.
 - **Disc Screens:** Consist of rotating discs on shafts, separating materials based on size and shape. They're ideal for dry, free-flowing materials like wood and plastics, preventing wrapping or clogging compared to trommel screens.
 - **Anti-Wrapping Screens:** Designed to prevent materials like plastics, cables, or fabrics from wrapping around rotating components, ensuring smooth operation. They use specialized designs (e.g., star screens) to handle fibrous or flexible debris.
5. **Manual Sorting Conveyors:**
 - Allow human workers to manually inspect and sort C&D waste, particularly for materials that automated systems struggle with.

- Conveyors move waste past workers stationed along the line, who remove contaminants, hazardous materials, or specific items (e.g., wood, plastics) that machines might miss. This step is vital for quality control and ensuring high-purity recycled products.

6. Air Drum Separators:

- Separate lightweight materials from heavier ones using air currents, improving material purity.
- Rotating drums with air jets blow lightweight debris (e.g., paper, plastics, wood) upward or away, while heavier materials (e.g., concrete, metals) fall to the bottom for further processing. This enhances sorting efficiency for mixed C&D waste.

7. Windsifters:

- Use air currents to separate materials by weight and density, similar to air drum separators but often for finer or lighter debris.
- Windsifters blow air through falling or conveyed waste, separating lightweight materials (e.g., plastics, paper) from heavier ones (e.g., concrete, metals). They're used in combination with other equipment for final sorting stages.

8. Magnetic Separators:

- Remove ferrous metals (e.g., steel, iron) from C&D waste, facilitating recycling and preventing equipment damage.
- Mounted on conveyors or over shredders, these use powerful magnets to attract and extract metal objects, which are then collected for recycling. Non-ferrous metals require additional separation with eddy current separators.

9. Sorting Robots:

- Automate the sorting process, increasing efficiency and accuracy for complex C&D waste streams.
- Equipped with sensors (e.g., cameras, infrared, or near-infrared) and AI, sorting robots identify and pick specific materials (e.g., plastics, wood) from conveyors, reducing reliance on manual labor. They handle high volumes and improve purity for recycling.

These components work together in a sequential or parallel workflow:

- **Cranes and feeders** initiate the process, moving and regulating waste flow.
- **Size reduction equipment** (primary and secondary shredders) breaks down materials for handling.
- **Screening systems** and **air drum separators/windsifters** separate materials by size and weight.
- **Magnetic separators** and **sorting robots** target specific materials like metals, while **manual sorting conveyors** address nuances automation misses.

TECHNOLOGY OF C&D WASTE PROCESSING

Construction and Demolition (C&D) waste processing technologies are crucial for managing waste sustainably, reducing landfill use, and promoting recycling. These technologies vary based on the scale of waste generation, from large urban centres to small towns.

C&D waste processing involves stationary units for large-scale waste, mobile units for flexibility, and mini or cluster approaches for small towns. The stationary units use crushing, sieving, and washing machinery, while mobile units offer on-site processing to save costs. The following table 4 summarizes the key features of each technology option.

Table-4: Features of each technology option.

Technology	Capacity (TPD)	Suitability	Advantages	Challenges
Stationary Processing	100–2,000+	Large cities, high waste volumes	High capacity, automated, comprehensive	High cost, fixed location, large space
Mobile Crushing Units	Up to 1,000	Large projects, remote sites	Flexible, cost-saving, on-site processing	Limited capacity, operational challenges
Mini Mobile Crusher	~5	Small towns, <20–30 TPD	Low cost, mobile, local management	Limited capacity, maintenance costs
Cluster Processing	Varies by cluster	Multiple small towns, shared facility	Economies of scale, regional cooperation	Coordination, cost-sharing, logistics

C&D MANAGEMENT (BUILDINGS)

On-site C&D waste management

Every construction project should earmark space on the project site for storage and segregation of waste. The size of the space must be determined based on waste generation estimate for the project. This can be made part of building permit requirement.

Project managers must be encouraged to reduce waste generation and generated waste should preferably be reused or/and recycled on-site. Depending upon the size of the project, the composition and characteristics of the waste multiple in-situ options are available. Some of the options are:

- **Backfilling**—The most common use for C&D waste in India is as a backfilling material. It is dumped in pits, trenches etc. and compacted for backfilling or used to increase elevation or make a top layer of surface for construction.
- **Direct reuse of valuables**—Materials of high reuse value like wood frames, unbroken bricks and ceramics are already widely recovered and sold in the secondary market. These are typically used for construction of temporary structures but if treated properly could be used for permanent structures as well.
- **Other applications**—C&D waste is also used in other minor applications as in filler material in roofing construction, wall decorative chips, etc.

Waste that cannot be utilized on site, must be stored in a segregated fashion, and transported to the city's disposal/recycling facility as prescribed by the city authority.

Transportation and collection of the waste

Waste from construction and demolition (C&D) should be transported to designated locations by generators or through other systems provided by Urban Landfills (ULLs), as per the by-laws. Both the generator and transporting entity must maintain records of the amount of waste transported to the designated dumping area. Vehicles carrying C&D waste should be covered to avoid dust, air pollution.

The transportation options are **door-to-door collection**, **intermediate collection points**, and **dedicated container systems**. Door-to-door collection services are already available in cities like Kolkata and Chennai, while intermediate collection points are being used in Delhi and Ahmedabad. Dedicated container systems allow municipalities to provide containers to builders for waste storage,

which can be picked up by the ULB for disposal or transferred to a ULB by generator based on the city's by-law.

ULLs can transport C&D waste on their own or hire private contractors, with transporters paid according to a schedule developed by the ULB. The ULB charges generators for transportation and disposal costs, and if there is a contracted C&D waste processing facility, the concessionaire may be paid a combined fee for collection, transportation, and processing per ton.

Collection points: Waste disposal practices in India involve the establishment of collection points, which are essential for efficient management and transportation. For large cities with limited land plots, small collection sites are recommended. The area required for transfer stations and processing facilities depends on the waste generated and the size of plots available, which needs to be determined by a feasibility study by the ULB. For example, Delhi has 168 C&D collection points/transfer stations and Ahmedabad has 16 designated collection points.

Disposal site selection: The disposal site selection is crucial, as a significant portion of waste is disposed of in low-lying areas, open spaces, and roadsides in an unauthorized manner. Changes should be made to prevent C&D waste from being dumped in landfills before recovering useful materials from the waste stream. The small fraction of C&D waste that comes out as unusable waste product after processing should be properly disposed of in a sanitary landfill and not mixed with other MSW. The hazardous fraction of C&D waste should be dumped in a hazardous waste landfill.

Specifications for setting up a recycling plant

More than 90% of the C&D waste composition in Indian cities can be processed or recycled and reused as secondary raw materials. Even for cities without dedicated recycling facilities, the C&D waste debris can be used for approved public works construction projects where possible, and the rest should be disposed at designated dumping sites for future recycling. Eco-sensitive areas like wetlands should not be used as dumping sites.

Setting up a recycling plant requires the state government or the concerned ULB to identify the land needed for processing facilities. The land should have adequate space for storage and sorting of incoming C&D waste, weighbridge, processing equipment, manufacturing and storage of finished products, testing, and administrative facilities. Off-site facilities such as access roads and electrical/water supply should also be considered.

Cities with high waste generation capacity may need multiple processing facilities in different zones, while cities with a generation capacity of 1,000 TPD or less can have a single centralized processing facility. Mobile crushing units may be suitable for processing C&D waste in small towns with low generation.

C&D MANAGEMENT (INFRASTRUCTURE)

C&D waste management for infrastructure projects is similar to that of building projects. Given that C&D waste generated from infrastructure projects like metro and flyovers is highly uniform and largely concrete by composition. It makes economic sense for mega developers like Delhi Metro Rail Corporation to have their own recycling facility as they can save on transportation cost and processing fee to ULBs, and custom engineer the recycled product to their project requirements.

C&D WASTE PRODUCTS

The major applications of C&D waste are listed below:

- **Granular Sub-Base (GSB)**—Crushed C&D waste can be used as GSB layer for road construction, regardless of the type of construction. The granular sub-base layer is formed by piling and compacting C&D aggregates of different sizes one over the other directly below the pavement surface. This acts as the load bearing and strengthening component of the pavement structure; in addition, it provides drainage for the pavement structure and protects the structure from frost.
- **Recycled Concrete Aggregates (RCA)**—Pure concrete waste can be recycled to make aggregates of different standard sizes to replace natural aggregates in construction processes.
- **Recycled Aggregates (RA)**—Crushed aggregates of standard size made from a mix of C&D waste materials is termed as Recycled Aggregates (RA). RA can be used for partial replacement of natural aggregates for construction of non-load bearing structures.
- **Manufactured Sand (M-Sand)**—Manufactured sand is also produced by crushing of C&D waste, and the finer particle fraction can be used to replace natural sand in construction of non-load bearing structures.
- **Smelting**—Scrap metal recovered from C&D waste is melted through smelting process and recycled to make new products.

Market uptake of C&D products:

- **Lack of Confidence in Recycled Products:** The market for recycled products from C&D waste is weak in India. Potential buyers are hesitant due to perceived inferior quality.
- **Poor Economic Viability of Recycled Products:** Currently, aggregates and manufactured products are taxed at 5% and 18% respectively, making the use of recycled products economically unviable.
- **GST relaxation for C&D waste recycled products,** including manufactured products, may be considered.

To ensure enhanced market uptake of C&D waste-based products, two approaches that can be adopted:

1. Preferential buy back by government and private entities
2. Eco-labelling and green certification of products

Examples of recycled C&D waste products used in India:

New Supreme Court complex: This extension project has used pre-cast cement concrete blocks of Grade M10 made of recycled C&D waste sourced from Delhi's Burari plant. About 1.8 million blocks were used as non-load bearing walls in the project.

Express Road in north Delhi: The DDA venture for construction of the pilot 100-m wide and 10-km long express road connecting NH-10 and Bakkarwala is using recycled aggregates of size –150mm to +26.5mm. About 600,000 tonnes of recycled C&D waste will be used to make the pilot road.

Initiatives in India Prior to C&D Rules

- **Mumbai:** A decentralized debris management solution promoted by YUVA and CIDCO in 1999 resulted in over 1,500 tons of waste recycling.
- **Delhi:** A pilot project developed by IL&FS Environmental Infrastructure & Services Ltd and the Municipal Corporation of Delhi in 2009 demonstrated the potential of collecting and recycling C&D waste.
- The pilot plant collected C&D waste from three designated zones of Delhi, converting it into Ready Mix Concrete, pavement blocks, kerb stones, and concrete bricks.

- Despite the success, the products manufactured by the recycling plant faced lack of acceptance due to lack of information and support from national legal instruments.

The Benefits of Construction and Demolition Recycling

Recycling C&D materials can save companies money by removing them from the site and reusing them at disposal plants or specialized facilities. The cost of recycling depends on the condition, amount, and type of materials. If the materials are in good condition, it is cheaper to send them to be recycled than to a landfill. Some state governments may provide tax benefits for recycling C&D items, but these require certain requirements.

Recycled C&D materials are typically cheaper to produce and purchase than new ones, as they are just as high-quality as new ones. Construction costs have increased an average of 19% over the last few years due to supply chain issues. Using quality recycled materials can increase revenue and create jobs.

C&D recycling is an example of a circular economy in action. It helps to create a closed-loop system where materials are reused and recycled, rather than being discarded.

Recycling construction and demolition (C&D) waste has many benefits, including:

- **Environmental protection:** Reduces the need for extracting new raw materials, which helps to reduce environmental degradation. It also reduces carbon emissions associated with extracting and manufacturing new materials.
- **Conserving resources:** Reduces reliance on finite natural resources like sand, phosphate, and fossil fuels.
- **Conserving landfill space:** Reduces the need for new landfills and associated costs.
- **Economic benefits:** Creates jobs in recycling industries and supports local communities. It also reduces building project expenses.
- **Sustainability:** Promotes sustainability by diverting waste away from landfills.
- **Public health and safety:** Enhances public health and safety.
- **Architectural style:** Helps to maintain the architectural style of an area by reintegrating locally sourced materials.
- C&D waste can be put to a profitable use, given the scarcity of sand and stone for construction, thereby saving natural resources.
- It prevents public nuisance and traffic congestion issues arising from indiscriminate dumping of C&D waste.
- It saves valuable space at landfill sites.
- It reduces cost of bulk transportation if recycled close to source of generation

Specific Material Benefits:

- **Concrete and Asphalt:**
 - Recycled aggregate: Crushed concrete and asphalt can be used as aggregate in new concrete mixes, road base, and fill material.
 - Reduces the need for virgin aggregate, which saves energy and reduces quarrying.
- **Wood:**
 - Reused timber: Salvaged wood can be reused for structural components, furniture, and decorative elements.

- Wood chips and mulch: Wood waste can be processed into wood chips for landscaping, mulch, and composite wood products.
- Reduces deforestation and the need for virgin timber.

- **Bricks and Masonry:**
 - Reused bricks: Salvaged bricks can be cleaned and reused in new construction or renovation projects.
 - Crushed brick aggregate: Crushed bricks can be used as aggregate in road base and fill material.
 - Preserves the embodied energy of the bricks.

- **Metals (Steel, Aluminium, Copper):**
 - High recyclability: Metals can be recycled repeatedly without losing their properties.
 - Energy savings: Recycling metals requires significantly less energy than producing them from raw materials.
 - Reduces mining and the associated environmental impacts.

- **Glass:**
 - Recycled glass aggregate: Crushed glass can be used in road construction and as aggregate in concrete.
 - Glass cullet: Recycled glass can be melted down and used to make new glass products.
 - Reduces the need for virgin silica.

Recycled Aggregate: Recycled aggregate is a sustainable alternative to natural aggregates and may be used in making concrete for non-structural purpose. The RA shall be free from deleterious material, such as, organic content, vegetable matter, coal, clay lumps, external substances such as, soft fragments like pieces of plastics, paper etc. Percentage of replacement of natural aggregates by RA can be up to 20% for any type of plain concrete work. The percentage can be increased up to 30% for road sub-base / base / other road related applications except wearing course.

They are made by crushing and processing waste materials like concrete, asphalt, and glass as shown in figure 5 and mentioned below:

- **Recycled concrete:** A solution to the problems of quarrying natural aggregates and disposing of old concrete.
- **Glass:** Can be used as a substitute for conventional materials in concrete mixes
- **Plastics:** Can be used as a partial replacement for fine aggregate in concrete
- **Ceramic waste:** Can be used to produce concrete with high freeze-thaw resistance and abrasion resistance



Fig5- Recycling process of C&D waste into aggregate

The table 5 shows the summary of various benefits of different wastes which can be incorporated as aggregates in concrete. The overall benefits of using recycled aggregates

- Recycled aggregates are cheaper than quarried stone.
- Using recycled aggregates reduces the need for mining, which helps preserve natural environments.
- Recycled concrete aggregates are durable and suitable for soft or wet ground.
- Recycled concrete aggregates use less energy to produce than traditional concrete.

Table -5: Summary of various benefits of different wastes which can be incorporated as aggregates in concrete.

Material	Benefits
Glass	Pozzolanic in nature, high thermal conductivity, reduced shrinkage, improved water absorption, reduced ecological emissions.
Plastics	Increased ductility, reduced shrinkage cracks, lightweight concrete.
Ceramics	Enhanced strength, required water absorption, low specific weight, and high pozzolanic nature.
Rubber	Protection against high temperatures and increase in strength.
Concrete	Pozzolanic in nature, high thermal conductivity, reduced shrinkage, improved water absorption, reduced ecological emissions.
Coir & Almond Wastes	Increased air content, improved mechanical strength and lower air density.

Recycling of supplementary cementitious materials (SCMs)

Using waste materials as supplementary cementitious materials (SCMs) in concrete can improve the concrete's durability, reduce costs, and reduce the carbon footprint of concrete production. SCMs are a cementitious addition to concrete that partially substitute Portland cement.

Biowaste can be converted into SCMs through thermochemical or mechanical processes, such as pyrolysis, gasification, combustion, and grinding. The table 6 represents the summary of various benefits of different wastes which can be incorporated as supplementary cementitious materials (SCMs) in concrete.

Table 6- Summary of various benefits of different wastes which can be incorporated as supplementary cementitious materials (SCMs) in concrete.

Material	Benefits
Metal Slag	High shear modulus, chemical stability, high strength.
Silica Fume	Pozzolanic nature, increased strength.
Rice Husk Ash	Enhanced compressive strength and improved water absorption.
Coal Ash/Fly Ash	Pozzolanic nature, good durability, low permeability, increased mechanical strength, reduced the alkali-silica reaction.
Ceramic Wastes	Increased strength, reduced permeability of concrete and increased efficiency.

Benefits of using SCMs

- SCMs improve the concrete's resistance to chemical attack and environmental degradation.
- SCMs can reduce the cost of concrete production.
- SCMs can make concrete more sustainable and eco-friendly.
- SCMs can help control the temperature of concrete, especially in mass concrete applications.
- SCMs can reduce the carbon footprint of concrete production because they often use industrial by-products.

Recycling construction and demolition (C&D) waste can have disadvantages, including environmental pollution, high costs, and low quality as mentioned below.

• **Sorting complexity:**

C&D waste often contains a wide variety of materials like concrete, wood, metal, drywall, and insulation, making it difficult and expensive to separate and sort for proper recycling.

• **Contamination concerns:**

If not properly handled, materials like paint, asbestos, or other hazardous substances present in C&D waste can contaminate the recycling stream, reducing the quality of recycled products.

• **Limited market for recycled materials:**

While some recycled C&D materials can be used in new construction, the market for these recycled products is often smaller compared to virgin materials, limiting the economic viability of recycling programs.

• **Transportation challenges:**

Due to the bulky nature of C&D debris, transporting it to recycling facilities can be costly and logistically complex, especially in urban areas.

- **Specialized equipment needs:**

Efficiently recycling C&D waste requires specialized processing equipment like crushers, shredders, and separators, which can be expensive to invest in and maintain.

- **Quality variations:**

Recycled C&D materials may not always meet the quality standards required for certain construction applications, potentially impacting the durability and performance of the final product.